

In the previous chapter, you studied in detail how nucleus of living cell is packed with different codes and information, necessary for the life processes. Scientists studied briefly DNA because the life, growth and unique features of an organism depend on its DNA. Scientists used several laboratory techniques to change the DNA of living organisms to produce desired qualities and characteristics resulted in the improved living standard for human being. The branch of science which deals with the use of microorganisms,

In this Chapter you will learn about:

- Biotechnology.
- DNA Replication.
- Introduction of Gene Bacterium.
- Genetic Modifications (Microorganism Resistance, Improved Nutrition and Quality of Food).
- Biotechnology Product Saving Lives (Insulin, Vaccines).
- General Applications (Agriculture, Environment, Health, Food Production and preservation).

All the students will be able to

- Define biotechnology.
- Explain how DNA is copied and made.
- Describe the relationship between DNA, genes and chromosomes.
- Define bacterium.
- Explain how genes are introduced into a bacterium.
- List some biotechnological products used in daily life.
- Explain that genetic modification in different foods can increase the amount of essential nutrients.
- List general application of biotechnology in various fields.
- Explain how biotechnology allows meeting the nutritional needs of growing populations.

animal cells, plant cells or their components to produce useful products for humans is called **BIOTECHNOLOGY**.

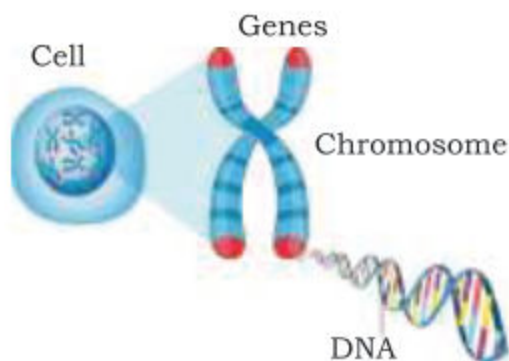


Fig.3.1: Gene, DNA, Chromosomes



Fig.3.2: Application of Biotechnology in Food Industry

BIOTECHNOLOGY

➤ Define Biotechnology

The word **biotechnology** is derived from two words, bio and technology. Bio means life and technology means scientific methods to create new products and solutions. Biotechnology is handling of living organisms or their components to perform practical task or to produce useful products.

Humans have practiced it to solve problems of food shortage and improve their way of life for several centuries. In past it was used to make alcohol and cheese by microorganisms, to do selective breeding. Selective breeding is a process used to develop new organisms with desirable characteristics of life stock and field crops, to produce antibiotics from microorganisms and to synthesize antibodies.

Now a days modern techniques have allowed us to manipulate genetic material of living organisms to develop a wide range of new products and improve the nutritional values of food products. Biotechnology is now playing a very important role in the diagnosis of infections as well as genetic disease. To understand biotechnology, let us first explore how the genetic material Deoxyribonucleic acid (DNA) replicates.



Fig.3.3: Bio plastics – a product which is degradable unlike synthetic plastic

DNA REPLICATION

- Explain how DNA is copied and made.
- Describe the relationship between DNA, genes and chromosomes.

In the previous chapters you have studied that DNA is the hereditary material. It is a very complex molecule. The units of this molecule are called nucleotides. Each nucleotide itself is made up of three components. They are;

- (i) Deoxyribose Sugar.
- (ii) Phosphoric acid.
- (iii) Organic basis.

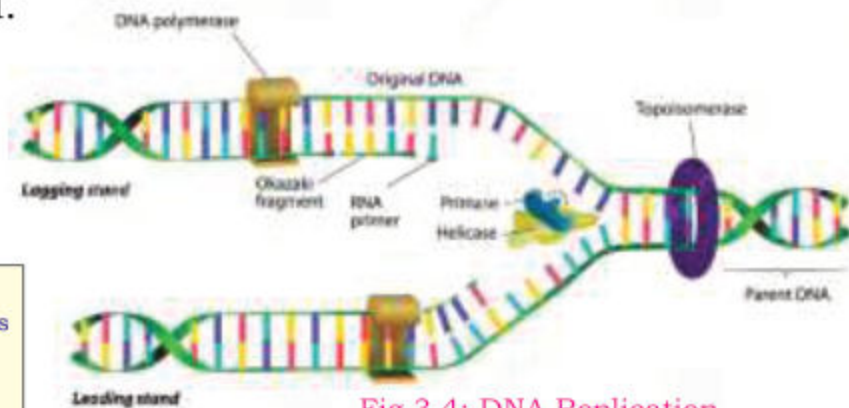
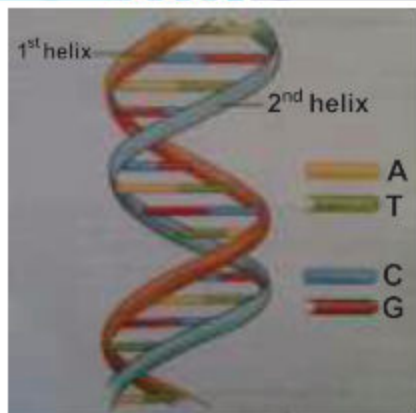
The information in DNA is stored in the form of code made up of four organic bases; Adenine (A), Guanine (G), Cytosine (C) and Thymine (T). Before cell division, the DNA material must be duplicated so that after all division, each new cell contains the full amount of DNA material. The process of making copies of DNA molecules within the cell nucleus is called DNA replication.

In 1953, James Watson and Francis Crick proposed the molecular model of DNA, which suggests the basic mechanism of DNA replication.

DNA Replication Process

DNA replication is the biological process of producing two identical copies of DNA from one original DNA molecule. This process occurs in all living organisms and is the basis for biological inheritance.

DNA is made up of a double helix of two complementary strands. During replication, these strands are separated. Each strand of the original DNA molecule then serves as a template for the production of its counterpart, a process referred to as semiconservative replication. As a result, the new helix will be composed of an original DNA strand as well as a newly synthesized strand.



Note: Teacher should encourage and facilitate students watching the video on a smart phone from the link.

Fig.3.4: DNA Replication

Genes, Chromosomes and DNA

As you know that DNA is the hereditary material present inside the nucleus of all cells. Each strand of DNA is called chromosomes. Gene is a distinct sequence of nucleotides forming part of a chromosome. It is a unit of heredity which is transferred from a parent to offspring and to determine some characteristic of the offspring.

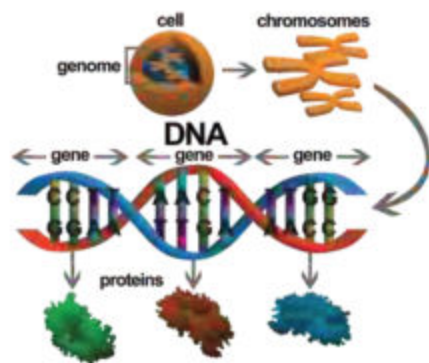


Fig.3.5: Gene, Chromosome and DNA

INTRODUCTION OF GENE INTO BACTERIUM

- Define bacterium.
- Explain how genes are introduced into a bacterium.

Bacteria are considered as the smallest and simplest living organisms. The bacteria cell does not have an organized nucleus. The DNA is found floating in the cytoplasm with other cell organelles. This single large circular strand of DNA is containing most of the genes needed for cell growth, survival, and reproduction. This chromosomal DNA tends to look like a mess of string in the middle of the cell. In addition to a single large piece of chromosomal DNA, cells also contain small pieces of DNA called plasmids. These plasmids (circular rings of DNA) are replicated independent of the chromosome.

Genetic engineering usually utilizes bacterial cells and their plasmids. To get a desired set of genes, scientists select and isolate the genes from one organism and insert it into the DNA of other organism which in this case is bacteria. For example, human insulin is made by isolating the gene of human insulin and then inserting this gene into the DNA of bacteria. These bacteria multiply; the new cells will contain the copies of the “engineered” plasmid. The foreign gene directs the cell to produce the human protein, insulin.

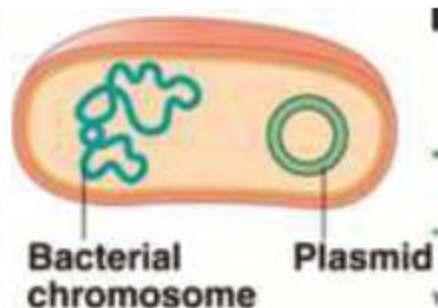


Fig.3.6: A Bacterium

GENETIC MODIFICATION

- Explain that genetic modification in different foods can increase the amounts of essential nutrients.
- Explain how biotechnology allows meeting the nutritional needs of growing populations.

Scientists concerned with feeding the growing human population are using DNA technology to improve the productivity of plants and animals important to agriculture. The scientists in agriculture field have already provided several crop plants with genes of desirable qualities. Biotechnology has played a revolutionary role in improving our agriculture and production of high yields of crops. Cultivation of such genetically modified crops improves the quality of crops and makes them safe for human consumption. The major crops that have been modified are maize, wheat, rice, canola potato, soya bean, cotton etc.

DO YOU KNOW?

In some poor countries, main food of people is rice. In such countries population faces an issue of malnourishment and deficiency of various key nutrients. One of the nutrients is Vitamin A, whose deficiency may cause early blindness and weak immune system among children. Gene of Vitamin A is inserted through biotechnology in the rice plant, creating a genetically modified rice variety.

BIOTECHNOLOGY PRODUCT SAVING LIVES (INSULIN, VACCINES)

- List some biotechnological products used in daily life.

Production of **Human insulin** is a great achievement of medical sciences. The human gene that secreted insulin is isolated from the pancreatic cell. The gene is inserted into the plasmid of bacterium. The recombinant bacteria with insulin gene are reproduced and the desired protein (insulin) is produced for commercial purpose. Similarly, Vaccines are the substances which contain the disease producing pathogens in the weakened form. When these vaccines are inserted into human body then the white blood cell produce special type of proteins (antibodies)

against these foreign particles to attain immunity against that disease. Scientists now use microorganisms to synthesize the desired vaccines. They identify the proteins of these diseases causing microorganism. These proteins when inserted in human body do not cause disease but stimulate the defense mechanism by producing the antibodies against that disease. This way certain life-threatening diseases such as Tuberculosis (TB), measles, typhoid, polio can be prevented when people are given the shots of these vaccines at an early stage.

DO YOU KNOW?

- Infants should be immunized with BCG vaccines against tuberculosis.
- MMR vaccines are given to children to develop resistance against measles.
- Typhoid vaccine is to be used against typhoid disease.
- Children can be immunized against polio disease with polio vaccine.

Activity 3.1:

Cut out news articles on outbreaks of disease in your country or other parts of the world. Find out what type of microorganisms, if any, cause this disease. How are these disease being treated or controlled? Are these treatments and controls effective? Discuss the articles in class.

GENERAL APPLICATIONS (AGRICULTURE, ENVIRONMENT, HEALTH, FOOD PRODUCTION AND PRESERVATION)

- List general application of biotechnology in various fields.

Biotechnology has made a revolution in the field of medicine, agriculture, environment and industries. Some of its common applications in our daily life are;

- It allows mass production of protein which was difficult in past.
- The production of human growth hormone by bio-technology is used to treat dwarfism.

- Insulin produced by bio-technology is being used to treat diabetics.
- Vaccines are made from treated bacteria or viruses.
- Cotton, corn, potato and soya bean plants have been engineered to be resistant to either insects or herbicides.
- It helps in enhancing the quality of crops like soya bean.

DO YOU KNOW?

In Pakistan, 80% of bananas are planted in lower Sindh. Sindh Agriculture University has introduced tissue culturing technique to increase the yield. The new variety of banana plant contains traits which delay quick ripening of fruit, thus helping in increased shelf life.



SUMMARY

- Biotechnology is the study of science that deals with the application of technology and materials on living organisms to derive a benefit in an ethical manner.
- DNA is the heredity material.
- Gene is the fundamental physical and functional unit of genetics.
- Genetic engineering is the scientific process of changing the genetic coding of an organism by inserting and replacing a section of a gene with the new one.
- Gene is basically responsible for producing certain kind of proteins which determines the physical and functional characteristics of an organism.
- Bacteria is used in genetic engineering because of its ability to multiply in shortest span of time.
- Gene replication is the process by which the DNA in a cell divides and replicates itself into two when cell division takes place.
- Biotechnology is now applied in the field of medicine, agriculture, environment and industries.
- Vaccines and insulin are everyday biotechnological products.

EXERCISE

Give short answer of the following

1. Define biotechnology.
2. Give some examples of achievements made in food and agriculture using biotechnology.
3. Describe role of Bacteria in biotechnology.
4. Write down some applications of biotechnology in daily life.

Choose the best answers:

1. In which organism, the human insulin is inserted for the commercial preparation of insulin:
a) Virus b) Bacteria.
c) Algae. d) Fungus.
2. The production of human growth hormone is used for the treatment of:
a) Night blindness. b) Dwarfism.
c) Osteomalaysia. d) Diabetes.
3. The process of making copies of DNA is called:
a) Lengthening. b) Regeneration.
c) Replication. d) Reproduction.
4. Name the vitamin whose gene is inserted in the rice plant creating a genetically modified rice variety:
a) vitamin B. b) vitamin K.
c) vitamin A. d) vitamin C.
5. Which of the following disease can be treated by the use of vaccine:
a) diabetes. b) measles.
c) AIDS. d) cancer.

6. The information in the DNA is stored in the form of code having _____ organic bases:
- a) 4 b) 5 c) 6 d) 2
7. Small pieces of DNA are called:
- a) Plasmids. b) replica. c) Template. d) Clone.
8. The scientists who proposed the DNA model in 1953 were:
- a) Haber and Bosch. b) Schleiden and Schwana.
c) Watson and Crick. d) Darwin and Crick.
9. Which of the following substances contains disease producing pathogens in the weakened form?
- a) Antibodies. b) Plasmids. c) Vaccines. d) Antibiotics.
10. Which of the following is not the organic bases of DNA?
- a) Guanine. b) Thymine.
c) Adenine. d) Uracil.

Match the column (A) and (B)

Column A	Column B
Tissue culturing technique	Double helix structure
Isolation of insulin	Structure of DNA
DNA	Copies
Cytosine and ribose sugar	Human pancreas
Replica	Increase of banana yield